

# Sec. 5 Experiments

## 1 EXPERIMENTS

**Instances.** *Adult* income classification [33]: logistic regression ( $d=109$  with intercept, cross-entropy,  $\eta=0.1$ ) on 100 race-homogeneous groups of 30 samples, the number of groups per race proportional to its prevalence (White/Black/Asian-Pac./Amer-Ind./Other = 85/10/3/1/1); the importance target is the fairness target itself, 1/5 per race divided equally among its groups, so a group from the smallest race carries  $p_i = 0.2$ . *IMDb-Wiki* age regression [34]: linear regression of age on 128-d ResNet face embeddings ( $\eta=10^{-2}$ ) over the first 100 identities with 30 images, with the cubic importance target  $p_i \propto (101 - i)^3$ . Observation rates are swept through a one-parameter family: on *Adult*  $r \propto p^\beta$ ,  $\beta \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$ , from uniform observation (prevalence regime,  $\nu=.60$ : the five groups of the three smallest races hold 60% of the target against  $r_i \approx K/m$ ) to importance-aligned ( $\nu=0$ ); on *IMDb-Wiki*  $r_i \propto i^\beta$ ,  $\beta \in \{0, \frac{1}{2}, 1, \frac{3}{2}, 2, 3\}$ , from uniform observation ( $\nu=.31$ , nine groups) to the mirror image of  $p$  ( $\nu=.88$ , 41 groups), plus the aligned  $r_i \propto 101 - i$  ( $\nu=0$ ). Eleven instances,  $\nu$  from 0 to .88.

**Protocol.** Each step observes  $K=3$  groups drawn without replacement with probabilities  $\propto r$ , which induces  $q$  over the  $\binom{100}{3}$  compositions;  $q$  is estimated from  $10^6$  draws of the sampler and the plan from (7) on the membership mask, once per instance. Each observed group takes  $H=5$  local steps on its own data before the weighted average, the standard practical deviation from the one-step analysis, applied to every rule alike since all rules consume the same local updates from the same draw  $A^t$ . Rules: *GAVOT*; *group-blind averaging*, the uniform average over the observed groups (Proposition 2 with  $\tilde{p}_i = r_i/K$ ); the *fixed multiplier*  $m/K$ , the rescaling  $(m/K)p_i$  that presumes uniform observation; and *full*, every group every step weighted by  $p$ , the reference  $\Phi(p)$ . Stepsize  $\eta_t = \eta/(1 + t/1000)$  unless stated, fixed before the comparison and shared by every rule; across nine schedules (constant, hyperbolic with  $T_0 \in \{250, \dots, 16000\}$ , inverse square root, step, cosine) the winner never changes on any instance, only the margin does: a faster decay puts full coverage on its floor and widens the IMDb-Wiki gap (cosine: 84.12 against 90.69), a constant stepsize narrows it to a tie. 4000 steps, 5 seeds; we report  $F_p$  averaged over the last 500 steps, mean  $\pm$  std over seeds. On both instances  $\arg \min F_w$  is available to machine precision (closed form for least squares, Newton for the logistic loss), so the floors  $\Phi(p)$ ,  $\Phi(\tilde{p})$ ,  $\Phi(\hat{p})$  of Theorem 8 are exact rather than estimated, and in every instance with  $\nu = 0$  the iteration (7) converges to a row error below  $3 \cdot 10^{-8}$ , certifying the remaining inequalities of (5).

**The least represented group.** Fig. 2 reads the same runs on the group the target exists for. On *Adult*, transport lowers the loss on race Other at every skew and under both stepsizes: .073 against .119 for group-blind averaging on the prevalence instance (full coverage .031), and .031 against .035 on the aligned one, where it matches full coverage. Amer-Indian, the other single-group race, behaves

Table 1: Headline cells, tail-500 over 5 seeds, decaying stepsize. *full* sees every group every step; *upsamp.*, self-normalized upsampling (weights  $\propto p_i/r_i$  within the batch); LDS, label-distribution smoothing [?], regression only. The fixed multiplier  $m/K$  diverges whenever observation rates are non-uniform.

| Instance         | GAVOT         | unif. avg | upsamp. | LDS   | $m/K$ | full   |
|------------------|---------------|-----------|---------|-------|-------|--------|
| Adult, $\nu=.60$ | <b>0.2269</b> | 0.2479    | 0.2288  | —     | 0.670 | 0.2073 |
| Adult, $\nu=0$   | 0.2075        | 0.2077    | 0.2075  | —     | div.  | 0.2073 |
| IMDb, $\nu=.31$  | <b>86.06</b>  | 91.63     | 86.74   | 94.28 | 8504  | 82.95  |
| IMDb, $\nu=0$    | <b>84.28</b>  | 86.37     | 84.40   | 89.43 | div.  | 82.95  |

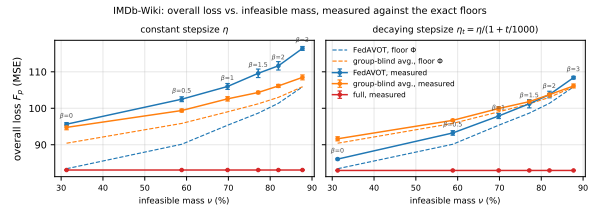


Figure 1: IMDb-Wiki severity sweep. Solid, measured tail-500 ( $\pm 1$  std); dashed, the exact floor  $\Phi(\cdot)$  each rule converges to. Left, constant stepsize: the optimization residual swamps the alignment gain and transport loses everywhere, even where the floors are seven units apart. Right, decaying stepsize: the residual collapses and the measured curves track their floors, so transport wins until the floors close at high  $\nu$ . The sign is set by which of the two terms of (9) dominates, not by the regime label.

the same way. On IMDb-Wiki the top-importance identities show the crossover of Table 2 one step earlier: under decay transport wins at  $\nu \leq .70$  (77.0 against 86.4 at  $\nu=.31$ , full 72.2; 73.6 against 79.9 when aligned) and loses at the three most infeasible skews, while at constant stepsize it wins only at  $\nu=.31$ . The single most starved group, the one with the largest  $p_i/r_i$  in each instance, sharpens the picture: on *Adult* it is race Other itself ( $p_i/r_i=6.7$ ); on IMDb-Wiki it is the top identity, where under decay transport gives 253 against 331 at  $\nu=.31$  and 373 against 380 at  $\nu=.59$  (full 219), then loses from  $\nu=.70$  on, the same crossover one step earlier still (a 30-image group, so  $\pm 5$  to 10 across seeds). The overall objective and the rare-group loss therefore tell the same story, with the rare group the more sensitive of the two.

**Why transport stops helping.** The failure is not a bug of the weights; it is the two terms of (9) changing sign. A starved group has  $p_i > r_i$ , so the plan can deliver at most  $r_i$  of its mass: past the coverage law of Cor. ?? the reachable target  $\hat{p}$  truncates at the observation rate, and as  $\nu$  grows  $\hat{p}$  and the group-blind  $\tilde{p}$  starve the same groups, so the two floors converge (Table 2,  $\Delta_{\text{floor}}$  from 6.99 to 0.02). What transport still does is concentrate weight on the starved group whenever it does appear,  $p_i/r_i$  times its batch share,

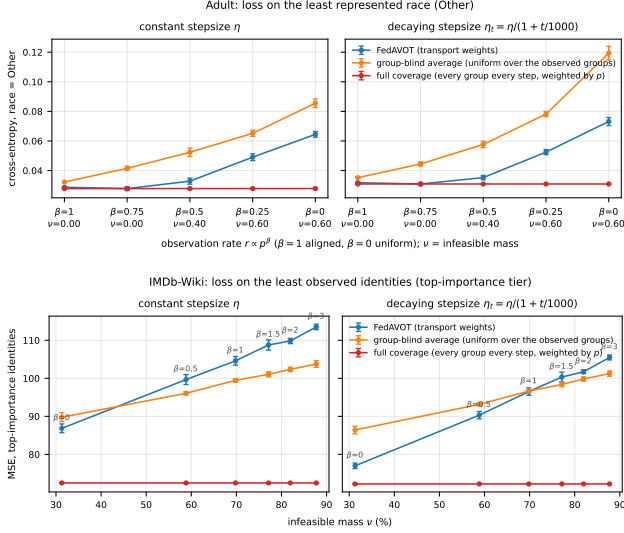


Figure 2: Loss on the least represented group, the metric the fairness target is about. Top, Adult: cross-entropy on race Other, one group of 30 out of 100, to which the category-uniform target assigns  $p_i = 0.2$ ; the axis is the observation-rate exponent,  $\beta=1$  aligned,  $\beta=0$  uniform (prevalence). Bottom, IMDb-Wiki: MSE on the 20 highest-importance identities, the least observed ones once  $\beta > 0$ , against  $\nu$ . Left panels constant stepsize, right panels decaying. Same protocol as the tables ( $K=3$ ,  $H=5$ , 4000 steps, tail-500 mean  $\pm$  std over 5 seeds); lower is better. Curves are the group-wise entries of the same runs as Table 1 and Fig. 1.

which is exactly the variance Lemma 6 does not see: the second moment is bounded, the variance about the mean is not, and it is largest on the rare group. Once the floors coincide that variance is all that separates the two rules, so transport loses by the residual, by 2 to 8 at constant stepsize and by 2.3 at  $\nu=.88$  even under decay. Transport helps exactly while there is mass left to move.

**E1: the headline cells.** Table 1 reports the four instances a practitioner would recognize. Transport wins on three: Adult under prevalence-proportional availability, 0.2269 against 0.2479 ( $t=45$ ), and both IMDb cells, 86.06 against 91.63 ( $t=21$ ) and 84.28 against 86.37 ( $t=13$ ). On Adult with importance-aligned availability the two coincide at the oracle value, which is what Proposition 2 predicts when  $\tilde{p} = p$ : there is nothing to correct and transport does not manufacture a difference. The fixed multiplier diverges on both aligned instances and is far off on the other two, so naive rescaling is no alternative to solving (4). Two imbalance baselines sit between the two. Self-normalized upsampling (weights  $\propto p_i/r_i$ , renormalized within the batch), the reweighting form of oversampling rare groups, recovers most of the gain and transport the rest: 0.2288 against 0.2269 on Adult prevalence ( $t=5.8$ ) and 86.74 against 86.06 on IMDb skewed ( $t=2.6$ , at the margin of significance with 5 seeds); on the aligned instances the two coincide, as every convex rule must. Downsampling ( $\min(p_i/r_i, 1)$ , normalized) is never better than upsampling (0.2308, 86.90). Label-distribution smoothing [?], the standard imbalanced-regression correc-

Table 2: IMDb-Wiki severity sweep, decaying stepsize.  $\Delta_{\text{floor}}$  is the alignment gain of Theorem 8, *gain* the measured advantage of transport,  $t$  its Welch statistic (population std). The excess residual transport pays averages 1.9, and the sign flips exactly where  $\Delta_{\text{floor}}$  crosses it. The five Adult skews all sit above their crossover and are omitted.

| $\beta$ | $\nu$ | $\Delta_{\text{floor}}$ | gain         | $t$  |
|---------|-------|-------------------------|--------------|------|
| 0       | .31   | 6.99                    | <b>+5.57</b> | 20.7 |
| 0.5     | .59   | 5.71                    | <b>+3.46</b> | 11.3 |
| 1.0     | .70   | 3.67                    | <b>+2.05</b> | 5.4  |
| 1.5     | .77   | 2.56                    | +0.58        | 1.1  |
| 2.0     | .82   | 1.62                    | −0.03        | −0.1 |
| 3.0     | .88   | 0.02                    | −2.28        | −7.9 |

tion, reweights by the rarity of the label value rather than by how rarely a group is observed, and trails even group-blind averaging on both IMDb-Wiki instances (94.28, 89.43). On the least represented group the margin of transport over up-sampling widens: 77.0 against 80.0 at  $\nu=.31$  and 73.6 against 75.6 when aligned on IMDb-Wiki, .073 against .075 on Adult prevalence. Upsampling is the one-shot heuristic for the marginal problem (4) solves exactly, without the feasibility certificate or the floor decomposition.

**E2: the crossover, and where the sign comes from.** Table 2 and Fig. 1 are the paper’s main evidence. On Adult, whose five skews are omitted from the table, the measured gain tracks  $\Delta_{\text{floor}}$  within 4% at three of five (.0134 against .0129, .0050 against .0049, .0002 against .0003) and exceeds it at the two most infeasible (.0210 against .0110, .0181 against .0144), group-blind averaging sitting further above its floor. On IMDb-Wiki the gain is uniformly below the floor by an excess optimization residual that is close to constant in  $\beta$ , mean 1.9 in MSE units, and the measured sign flips precisely where  $\Delta_{\text{floor}}$  crosses that value, between  $\beta=1.5$  (2.56, gain +0.58) and  $\beta=2.0$  (1.62, gain −0.03). This is Theorem 8 read at finite  $T$ : the sign of the comparison is a quantity one computes in advance.

**E3: the  $\ell_1$  bound is too loose to be the criterion.** The distances  $\|p - \tilde{p}\|_1$  and  $\|p - \tilde{p}\|_1$  are also computed on every instance, and  $\|p - \tilde{p}\|_1 < \|p - \tilde{p}\|_1$  holds in all eleven, including the three where transport loses or ties. The  $\ell_1$  form of Theorem 8 therefore never predicts a loss and cannot serve as the criterion; the exact floors can.

**E4: the residual is a stepsize artifact.** The left panel of Fig. 1 runs the identical instances at constant stepsize. There transport loses on all six IMDb skews, by as much as 7.9, even at  $\nu=.31$  where the floors are 6.99 apart. The transport weights  $Y[\cdot, j]$  concentrate mass on rarely observed groups, so although Lemma 6 bounds the second moment identically for both rules, the variance about the mean is larger, and at constant stepsize the iterates sit on a noise floor proportional to it. Under  $\eta_t = \eta/(1 + t/1000)$  that floor decays, the measured curves fall onto their dashed floors, and every sign reversal attributable to the stepsize disappears. The residual is a property of the schedule, not of the estimator.

**E5: the decomposition in closed form.** The mirrored IMDb-Wiki instance ( $\nu=.88$ ) is where Theorem 8 can be read off exactly. The Sinkhorn limit truncates the target at the observation-rate ceiling,  $\tilde{p}_i \approx \min(p_i, r_i)$  up to renormaliza-

tion (maximum deviation .023), with  $\|p - \hat{p}\|_1 = 1.58$  of a possible 2. The frontier is least squares, so  $\Phi(p) = 82.9$ , matching the measured full plateau 82.95, and  $\Phi(\hat{p}) = 105.9$ : the alignment floor is 23.0 and the realized bias equals its exact expression  $|(p - \hat{p})^\top L(\theta^\dagger)| = 30.7$ , where  $L$  collects the per-group risks, while the  $\ell_1$  form  $2M\|p - \hat{p}\|_1$  is conservative since  $M \geq \max_i L_i(\theta^\dagger) \approx 454$ . Group-blind averaging has the same floor to two decimals,  $\Phi(\tilde{p}) = 105.9$ : both surrogates starve the same groups, so  $\Delta_{\text{floor}}$  vanishes and the entire measured gap between the two rules at this endpoint of Table 2 is residual, which is why it is the one instance where transport still loses by a significant margin under decay ( $\beta=2$ , where the floors are 1.6 apart, is a tie).

**E6: risk aversion, and where it does not help.** Sweeping  $(\alpha, \gamma)$  on a  $10 \times 10$  grid, the overall objective is minimized on the  $\gamma = 1$  edge on every instance, which is  $\kappa(1, \gamma) = 1$  made visible and confirms that risk aversion costs the average. The one place the CVaR layer earns its keep is the worst race on Adult under prevalence availability, where GAVOT+CVaR at  $(0.2, 0.7)$  reaches 0.3624 against 0.3679 for GAVOT and 0.3994 for group-blind averaging. Per race, transport improves four of five (Black .179 vs .200, Asian-Pac .368 vs .399, Am-Ind .175 vs .189, Other .073 vs .119) and regresses only the majority group (.339 vs .332), which is what reweighting toward a category-uniform target is supposed to do.

**Why risk aversion does not reach the rare group.** On the least represented group the CVaR layer never helps: over the whole  $(\alpha, \gamma)$  grid the best rare-group loss is at  $\gamma=1$ , i.e. with the layer switched off, on all four headline instances, and at  $(\alpha, \gamma) = (0.1, 0.1)$  it nearly triples the loss on race Other (.207 against .073) and raises the top-identity loss on IMDb-Wiki from 77.0 to 103. The reason is where the tilt lands. CVaR upweights the groups whose realized loss exceeds the running threshold, but it can only tilt among the groups in the batch, and the rare group is absent from most batches: the extra weight goes to whichever observed group is doing badly that step, not to the group the target is about. Composed with transport, the hinge multiplies weights that are already  $p_i/r_i$  large, so the product amplifies the variance of (9) rather than the alignment, which is why the loss grows as  $\alpha$  shrinks. Risk aversion changes the functional; it does not change who is observed, and observation is what the rare group lacks. Transport moves mass toward it whenever it appears; CVaR cannot, and the one gain it produces, on the worst race above, comes from tilting among the groups that are present.